

Abel Tadesse

Software Engineer — Backend & Mobile Application Developer

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PROFESSIONAL SUMMARY

Software engineer with 3+ years of experience building secure, scalable financial platforms, REST APIs and cross-platform mobile applications. Led backend and mobile teams at a savings and credit cooperative, delivering self-service banking systems used in production and resolving security issues that enabled certification for integration with partner banks. Four years of remote mobile development experience and a Top Rated Upwork track record with a 100% Job Success Score across ML, backend and Flutter projects. Strong foundation in Django, FastAPI and Flutter, with growing expertise in applied machine learning through ongoing MSc studies. Part-time university lecturer with experience teaching enterprise and mobile application development, and growing interest in applied AI research for financial systems and human-computer interaction.

INDUSTRY EXPERIENCE

Mobile App Team Lead & Backend Developer

Feb. 2025 – Present

Amigos Saving and Credit Cooperative Society (SACCO)

Addis Ababa, Ethiopia

- Lead a cross-functional mobile team building Flutter-based financial applications, including responsive UI, input validation and loan planning tools for cooperative members.
- Architected and scaled Django/DRF backend APIs serving mobile and web clients, improving reliability across multiple integrated banking systems.
- Introduced code review processes that reduced post-deployment bugs by an estimated 30%, and mentored junior developers on secure API design and state management.
- Designed automation features for digital loan products, improving processing efficiency and client evaluation accuracy.

Backend Team Lead & Application Developer

Mar. 2024 – Feb. 2025

Amigos Saving and Credit Cooperative Society (SACCO)

Addis Ababa, Ethiopia

- Led development of systems processing thousands of daily transactions with role-based workflows and secure APIs.
- Resolved critical security vulnerabilities in the core banking API layer, enabling successful integration certification with partner banks.
- Built a self-service platform with JWT authentication, OTP verification and multi-role document management workflows, significantly reducing manual processing time.
- Optimised PostgreSQL queries and introduced Celery-based asynchronous task processing, improving throughput for high-volume financial operations.

Application Developer

Jul. 2023 – Mar. 2024

Amigos Saving and Credit Cooperative Society (SACCO)

Addis Ababa, Ethiopia

- Developed and maintained backend APIs and reporting services for financial products, ensuring data integrity and security across internal systems.
- Collaborated with the mobile team to design APIs tailored for Flutter applications and supported deployment of mobile features in production.

Freelance Software Engineer – Top Rated

2025 – Present

Upwork (Remote)

International clients

- Top Rated freelancer with a 100% Job Success Score, delivering machine learning, backend and Flutter projects for international clients, all rated 5.0/5.0.
- Built the Sheffield Crime Trends Visualiser for a UK client: an interactive crime analytics and forecasting dashboard using Python, machine learning and Plotly Dash.
- Delivered end-to-end projects including a production Flutter mobile app, ML model deployment with backend API integration, and Python desktop applications.

Mobile Application Developer (Remote)

2021 – 2025

Freelance client engagement

Remote

- Developed and maintained cross-platform Flutter applications for a remote client over a four-year engagement, owning the full lifecycle from requirements to release.
- Delivered iterative feature updates, bug fixes and performance improvements based on user feedback, sustaining a long-term client relationship.

TEACHING EXPERIENCE

Lecturer (Part-time)

2025 – Present

Addis Ababa University, School of Information Technology and Engineering

Addis Ababa, Ethiopia

- Teach *Enterprise Application Development* and *Mobile Application Development* to final-year software engineering students.
- Design and deliver course content on backend architecture, REST API design, authentication and authorization, and Flutter-based mobile development.
- Supervise senior projects focused on financial systems, mobile apps and applied machine learning integration into production applications.

EDUCATION

Addis Ababa University

Addis Ababa, Ethiopia

MSc in Artificial Intelligence (Ongoing) | GPA: 3.5/4.0

2024 – Present

- Focus areas: deep learning, computer vision, natural language processing and model deployment in production systems.
- Coursework includes advanced machine learning, reinforcement learning, and AI for decision support systems.

Addis Ababa University

Addis Ababa, Ethiopia

BSc in Software Engineering | GPA: 3.16/4.0

2019 – 2024

- Capstone: design and implementation of a full-stack enterprise application with secure backend and Flutter frontend.
- Completed projects in distributed systems, database design and mobile application development.

RESEARCH & INTERESTS

Applied machine learning for financial risk scoring, fraud detection and process automation.

Scalable backend architectures for high-volume transaction systems and real-time analytics.

Human-computer interaction for mobile financial applications and user-centric design in cooperative banking.

TECHNICAL SKILLS

Languages: Python, Dart, Java, SQL, Bash

Backend: Django, Django REST Framework, FastAPI, REST API Design, JWT/OAuth/OTP Authentication, RBAC

Mobile: Flutter, Provider State Management, Firebase, Supabase

Databases: PostgreSQL, MySQL, SQLite, Redis

DevOps & Tools: Docker, Git, CI/CD, Celery, Swagger, VPS Deployment

AI / ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, NLP, Model Integration

Other: System Design, Agile, Code Review, Technical Mentorship

SELECTED PROJECTS

SACCO Self-Service Backend | *Django, DRF, PostgreSQL, Celery, Redis, Docker*

2025

- Production API platform with OTP verification, JWT authentication, RBAC and client-specific document workflows; deployed via Docker with a CI/CD pipeline.

SACCO Loan Calculator App | *Flutter, Dart, Provider*

2025

- Cross-platform loan calculator with real-time amortisation schedules, dynamic UI and clean architecture principles, built for cooperative members.

Enterprise ERP Change Detection Model | *Python, PyTorch, NLP*

2025

- Custom NLP models detecting enterprise software transformation signals from job postings, websites and filings, with automated scraping pipelines across 100+ data sources.

Real-Time Emotion Detection System | *FastAPI, TensorFlow, OpenCV, React*

2025

- Real-time emotion classification using a Mini-Xception CNN, served through a FastAPI backend with live webcam streaming to a React frontend.

AWARDS & HONORS

Top Rated Upwork Freelancer status with 100% Job Success Score for sustained high client satisfaction.

AMIGOS SACCO Best Employee of the Year 2024/2025

PROFESSIONAL ACTIVITIES

Technical mentorship of junior developers in backend design, testing and code review practices.

Participation in local tech meetups and online communities focused on Python, Django and Flutter.

Occasional informal talks for students on freelancing, remote work and production-grade API development.

LANGUAGES

Amharic: Native | **English:** Fluent (professional working proficiency)

CERTIFICATIONS

Flutter Advanced Course – Clean Architecture with MVVM (*Udemy, Aug 2024*)

Python & Django REST API Bootcamp – Build A Python Web API (*Udemy, Jun 2024*)

Deep Learning Specialization (*Coursera, audited*)